

Selim Molla

PHD CANDIDATE ■ COMPUTATIONAL SCIENCE PROGRAM

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RESEARCH EXPERIENCE

University of Texas at El Paso, USA

September 2023 - Present

Graduate Research Assistant, Dept. of Industrial, Manufacturing, and Systems Engineering

- Conduct research at the “**Research Institute for Manufacturing and Engineering Systems (RIMES)**” in computational modeling, simulation, optimization, artificial intelligence, and data-driven decision-making.
- Develop simulation-based models using FlexSim, FlexSim HC, and AnyLogic for manufacturing, healthcare, and complex operational systems.
- Apply machine learning, deep learning, and reinforcement learning methods to forecasting, monitoring, optimization, and simulation-based decision-making problems.
- Integrate mathematical optimization with discrete-event simulation to support resource allocation, production planning, and system-performance analysis under uncertainty.
- Conduct simulation experiments, statistical and performance analysis, model validation, and research dissemination through peer-reviewed publications and conference presentations.

TEACHING EXPERIENCE

University of Texas at El Paso, USA

September 2026 - Present

Co-Instructor, Industrial Systems Simulation course, IMSE Department

- Co-teach the Industrial Systems Simulation course, including lectures, laboratory exercises, simulation modeling activities, and student support.
- Develop and demonstrate simulation models and guide students in applying discrete-event simulation concepts using FlexSim.

University of Texas at El Paso, USA

September 2023 - Present

Graduate Teaching Assistant, Computational Science Program

- Assist with undergraduate teaching through review sessions, grading, examination proctoring, and student support.
- **Courses assisted:** Probability and Statistics, Calculus I, Calculus III, Math for Social Sciences I, Matrix Algebra, Precalculus

Khulna University of Engineering & Technology, Bangladesh

September 2021 – December 2022

Teaching Assistant, Department of Industrial Engineering and Management

- Assisted with grading script and proctoring examinations

EDUCATION

Ph.D. in Computational Science

University of Texas at El Paso (UTEP), USA.

in Progress

M.Sc. in Computational Science

University of Texas at El Paso (UTEP), USA.

Awarded: 12/2025

CGPA: 3.94/4.00

B.Sc. in Industrial & Production Engineering

Khulna University of Engineering and Technology (KUET), Bangladesh.

Awarded: 05/2017

GPA: 3.29 /4.00

RESEARCH INTEREST

- Computational modeling, simulation, and optimization of large and complex systems
- Machine learning, deep learning, and reinforcement learning for manufacturing and healthcare applications
- Artificial intelligence and digital twins for intelligent manufacturing systems and decision support
- Data-driven methods for resource allocation, system optimization, and operational decision-making

RESEARCH PROJECTS

M.Sc. thesis title: *“Process-Centric Simulation to Optimize Complex Production Lines Considering Multi-Class Parts Combination.”*

- Developed a hybrid simulation-guided mathematical framework (SiGMa) integrating mathematical programming and discrete-event simulation to optimize workforce allocation in mixed-model assembly lines under real-world variability.
- Formulated a mathematical programming model incorporating system constraints and solved it using Python (PuLP).
- Built and validated a FlexSim-based discrete-event simulation (DES) model to verify optimization results and capture system dynamics.
- Applied the approach to an industrial case study, demonstrating improved workforce utilization and production efficiency.

B.Sc. thesis title: *“Line Balancing by Linear Programming.”*

- Formulated and solved an assembly line balancing problem using linear programming to optimize task allocation across workstations.
- Implemented multiple heuristic methods (Largest Candidate Rule, Kilbridge & Wester, Ranked Positional Weights) and compared their performance with the optimal LP solution.
- Developed Python-based code to automatically generate objective functions and constraints, and solved the model using LINDO.
- Demonstrated improved efficiency of linear programming over heuristic approaches in workload balancing and resource utilization.

PUBLICATIONS

Published

- **Molla, S.**, Rahman, M. F., Tseng, T. L. B., Chiou, R. Y., & Quddus, M. A. (2026). A Risk-Free Approach to Validate Drone Flight Paths Using Computer Simulation and Reinforcement Learning. In *Proceedings of the 2026 ASEE Annual Conference & Exposition*.
- Arvizu Astorga, J. F., **Molla, S.**, Tseng, T. L. B., & Rahman, M. F. (2026). A Noble Digital Twin-Based Inspection Scheme (DTIS) for Real-Time Monitoring in Smart Manufacturing. In *Proceedings of the Manufacturing Science and Engineering Conference (MSEC 2026)*.
- Siddique, M. A., **Molla, S.**, Lopes, A. J., & Rahman, M. F. (2025). Simulation-Based Job Shop Scheduling to Minimize Resource Uncertainties in Manufacturing Environment. In *Proceedings of the 2025 Winter Simulation Conference (WSC)*, 1541–1552.
- Rahman, S. M. A., **Molla, S.**, Sultana, J., Chiou, R. Y., Tseng, T. L. B., & Rahman, M. F. (2025). Development of Digital Laboratory Modules Using Computer Simulation for Enhanced Learning Experience in Manufacturing Education. *Manufacturing Letters*.
- **Molla, S.** (2025). *Process-Centric Simulation To Optimize Complex Production Lines Considering Multi-Class Parts Combination*. Open Access Theses & Dissertations, 4571.
- Hasan, M. R., **Molla, S.**, & Siddique, I. M. Next-Gen Production Excellence: A Deep Simulation Perspective on Process Improvement. *Journal of Mechatronics, Machine Design and Manufacturing*, 6(1), 7–20.
- Noman, A. H. M., Mustaquim, S. M., **Molla, S.**, & Siddique, I. M. (2024). Enhancing Operations Quality Improvement Through Advanced Data Analytics. *Journal of Computer Science Engineering and Software Testing*, 10(1), 1–14.
- Ullah, M. R., **Molla, S.**, Siddique, I. M., Siddique, A. A., & Abedin, M. M. (2023). Optimizing Performance: A Deep Dive Into Overall Equipment Effectiveness (OEE) for Operational Excellence. *Journal of Industrial Mechanics*, 8(3), 26–40.
- Siddique, I. M., Siddique, A. A., Smith, E. D., & **Molla, S.** (2023). Assessing the Sustainability of Bitcoin Mining: Comparative Review of Renewable Energy Sources. *Journal of Alternative and Renewable Energy Sources*, 10(1), 1–12.

Accepted Conference Paper

- **Molla, S.**, Raut, S., Hossain, N. U. I., Sokolov, A., Gorod, A., & Rahman, M. F. (2026). AI-Enabled Forecasting for Wind Energy Management. In *Proceedings of the American Society for Engineering Management (ASEM) 2026 International Annual Conference*.

Accepted Book Chapter

- Rahman, S. M. A., **Molla, S.**, & Rahman, M. F. (2026). *Systematic Data-Driven Approaches and Applications to Overcome the Risk of Supply Chain Disruption and Mitigation*. Book chapter accepted for publication by Taylor & Francis.

Under Review

- **Molla, S.**, Rahman, M. F., Rahman, S. M. A., Tseng, T.-L. (Bill), Lopes, A. J., & Hossain, N. U. I. (2026). A Simulation-Guided Optimization Framework for Mixed-Model Assembly Lines under Process Stochasticity. *European Journal of Operational Research*. Under review.
- Rahman, S. M. A., **Molla, S.**, Rahman, M. F., Tseng, T.-L. (Bill), Hossain, N. U. I., & Quddus, M. A. (2026). Intelligent Job-Shop Operations Using Digital Twin-Integrated Fallback Routing and Dynamic Dispatching. *Expert Systems with Applications*. Under review.
- **Molla, S.**, Rahman, M. F., & Hossain, N. U. I. (2026). Predicting Blockchain-Enabled Information Integration in Healthcare Supply Chain. *Smart Health*. Under review.

RESEARCH PRESENTATIONS

- **Technical Presentation**, ASME Manufacturing Science and Engineering Conference (MSEC 2026), The Pennsylvania State University, State College, PA, 2026.
- Poster Presentation, IMSE Day, 2026, UTEP
- Poster Presentation, IMSE Day, 2025, UTEP
- Poster Presentation, IMSE Day, 2024, UTEP

PEER REVIEW EXPERIENCE

- **Reviewer for *International Journal of Research in Industrial Engineering***
Reviewed 15 manuscripts (2024 – 2026)
- **Reviewer for *F1000Research* [Ad hoc]**
Reviewed 1 manuscript (2024)

TECHNICAL STRENGTHS

- **Simulation & Modeling:** FlexSim [advanced], FlexSim HC, and AnyLogic softwares
- **Machine Learning & AI:** Machine learning, deep learning, reinforcement learning, data analytics
- **Programming Language:** Python, MATLAB, C/C++, R
- **Programming Libraries & Tools:** NumPy, Pandas, Scikit-learn, Keras, TensorFlow
- **Statistical Software:** SPSS, Minitab
- **3D Modeling and Design:** AutoCAD, SolidWorks

THEORITICAL STRENGTHS

Introduction to computational science; Numerical analysis; Mathematical and Computer Modelling; Advanced scientific computing; Advanced Algorithms; Computational Methods of Linear Algebra; Computer Simulation Application; Industrial Data Analytics; Stochastic Processes; Machine Learning; Deep Learning; Time Series Analysis; Statistical Analysis; Deterministic Decision Models; Stochastic Decision Models; Combinatorial Optimization; Computer-integrated Manufacturing Systems; Modern Manufacturing Processes, Operations Research, Operations Management

HONORS & AWARDS

- NSF Student Researcher Travel Award in 2026
- Conference Travel Fund from the College of Science in 2026
- **2nd Place (Runner-Up)**, FlexSim Student Simulation Competition, Healthcare Systems Process Improvement (HSPI) Conference, 2024
- Technical Education Scholarship in KUET (2013 - 2016)
- Scholarship awarded by a Bengali community organization in Germany (2010–2016)
- Board scholarship for academic excellence in 2010, Jessore board, Bangladesh.

PROFESSIONAL SERVICES

Logistics Secretary, Bangladeshi Student Association (BSA), University of Texas at El Paso (UTEP)

2025–2026

- Support organizational logistics and coordination for academic, cultural, and community events.

Student Mentor, MARCS Tutoring Center, University of Texas at El Paso (UTEP)

Fall 2024 & Spring 2025

- Provided academic support and problem-solving assistance to students in mathematics-related subjects.

REFERENCES

- **Dr. Ming-Ying Leung**

Director

Computational Science and Bioinformatics Programs

University of Texas at El Paso

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- **Dr. Md Fashiar Rahman**

Assistant Professor

Department of Industrial, Manufacturing, and Systems Engineering

University of Texas at El Paso

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- **Dr. Tzu-Liang (Bill) Tseng**

Chair & Professor

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